BL Weather Report

Technical Guide

Rainfall Monitoring, Mapping & NDVI Analysis — Methodology & Calculation Reference

Version 2.0 · Generated August 13, 2026

1. Overview

The **BL Weather Report** workflow is a data-to-report pipeline built for Big Life Foundation's rainfall monitoring programme in the Amboseli ecosystem. It ingests rainfall-gauge observations from **EarthRanger**, derives vegetation condition per ranch from **Google Earth Engine** NDVI time-series, and delivers rainfall charts, an interactive map, a summary table, per-ranch NDVI trend charts, a Word report, and a dashboard.

The workflow is implemented as an **Ecoscope Workflow** (spec.yaml) and executes within the ecoscope-workflows runtime. NDVI analysis is fanned out per ranch using the mapvalues directive, grouping by the R_NAME column of the Amboseli conservancy boundaries GeoDataFrame.

> **Note:** This version of the guide (2.0) reflects a substantially expanded spec.yaml compared to the original release: an interactive rainfall map (land use, ranch boundaries, electric fence, clustered rainfall readings), a rainfall-by-station summary table, and a 4-widget dashboard have all been added. The previously documented per-ranch NDVI CSV export and dynamic “generated by” author lookup are no longer part of the pipeline.

The workflow delivers the following outputs per run:

Category	Outputs
Rainfall charts	Overall rainfall bar chart (per station), monthly stacked bar chart (per station × calendar month)
Summary table	Sortable, filterable rainfall-by-station table
Rainfall map	Interactive map: Amboseli land use, ranch boundaries, electric fence, and a clustered proportional-symbol layer of station readings
NDVI charts	One historic NDVI time-series chart per ranch, with a min/max/mean band
Data file	Aggregated station rainfall readings, with point geometry (GeoParquet)
Report	Word document (overall_report.docx): Big Life weather report template populated with every chart, the map, and all NDVI charts
Dashboard	4 widgets: rainfall map, overall rainfall chart, monthly rainfall chart, rainfall-by-station table

2. Dependencies & Prerequisites

2.1 EarthRanger Connection

All rainfall data is sourced from an **EarthRanger** instance via set_er_connection. The workflow fetches events of type rainfall_rep using get_events, with include_details: true so the event_details JSON blob carrying the actual gauge readings is returned. raise_on_empty: true is set, so the workflow raises an error rather than silently skipping if no rainfall events are found for the period — distinguishing a data-availability problem from a genuine zero-reading period.

2.2 Google Earth Engine Connection

A **Google Earth Engine (GEE)** service account is required for NDVI analysis. set_gee_connection stores the connection, which is passed to calculate_ndvi_range for each ranch ROI. GEE credentials must be stored as a JSON private key; see the GEE Setup Guide for instructions.

2.3 Static Geospatial Layers

Three GeoPackage files are downloaded automatically from Dropbox at run time via fetch_and_persist_file (3 retries, overwrite_existing: false so a cached copy is reused on subsequent runs). All three are reprojected to **EPSG:4326 (WGS 84)** before use.

File	Layer used for	Key columns
amboseli_ranch_conservancies_layers.gpkg	Land-use fill polygons on map	land_use (5 categories)
amboseli_group_ranch_boundaries_x_electric_fence.gpkg	Ranch boundaries + electric fence lines on map	land_use (Ranch boundaries, Electric Fence)
amboseli_group_ranch_boundaries.gpkg	ROI polygons for per-ranch NDVI calculation	R_NAME (ranch / conservancy name)

2.4 Base Map Tile Layers

Every map uses two stacked tile services, configured via set_base_maps_pydeck:

Layer	URL (abbreviated)	Opacity	Max zoom
ESRI World Hillshade	arcgisonline.com/…/World_Hillshade/…	1.0	20
ESRI World Street Map	arcgisonline.com/…/World_Street_Map/…	0.15	20

2.5 Grouping Strategy

The grouper is fixed to R_NAME — the ranch / conservancy name field in the Amboseli boundaries GeoPackage. This column drives the split_groups fan-out that produces one NDVI chart per ranch, and is also carried through to the temporal index added to rainfall events and to the final dashboard metadata.

3. Rainfall Data Pipeline

3.1 Event Fetch — rainfall_rep

get_events queries EarthRanger for events of type rainfall_rep within the analysis time range, with force_point_geometry: true and null-geometry events excluded. Updates and related events are not fetched (include_updates: false, include_related_events: false).

Columns retained per event:

- id, time, event_type, event_category, reported_by, serial_number, geometry, created_at, event_details

3.2 Detail Titles & JSON Normalisation

process_events_details (a Big Life-specific task) maps the raw event-detail keys to their human-readable question titles (map_to_titles: true) and preserves field order (ordered: true) before the JSON is flattened. normalize_json_column then flattens the event_details object into individual event_details__<title> columns (sort_columns: true); skip_if_not_exists: true makes the step a no-op if the column is already flat.

3.3 Column Renaming

map_columns renames the flattened detail columns to short working names and drops columns that are no longer needed:

Raw column name	Working name	Description
event_details__Digital rain gauge (mm)	digital_raingauge	Reading from the digital rain gauge
event_details__Manual rain gauge (mm)	manual_raingauge	Reading from the manual rain gauge
event_details__Rain gauge station	raingauge_station	Name of the gauge station
event_details__Rain gauge station zone	zone	Zone associated with the station

Dropped columns: event_type, event_category, reported_by, serial_number, patrols, event_type_display. raise_if_not_found: false means a missing source column is skipped rather than raising an error.

3.4 Combined Readings — Digital / Manual Fallback

combine_columns_with_fallback merges the digital and manual gauge readings into a single overall_readings column, in place:

- **Primary:** digital_raingauge — used when available and non-null
- **Fallback:** manual_raingauge — used when the digital reading is absent
- **fill_remaining: 0** — rows where both are null receive a zero reading

This design reflects field practice: digital gauges are preferred for accuracy, but rangers fall back to manual readings when the digital gauge malfunctions or has not yet been read.

3.5 Data Cleaning

convert_columns_to_float casts overall_readings to a float column; values that fail to parse are coerced to NaN (errors: coerce) and then filled with 0 (fill_value: 0), so the aggregation step below receives no nulls.

3.6 Aggregation by Station

spatial_summarize_df collapses all events to one row per gauge station (groupby_cols: [raingauge_station]):

Column	Aggregation	Result
geometry	first	Station point geometry (all readings for a station share the same location)
raingauge_readings	sum	Total rainfall in mm at each station for the period (1 decimal place); source column: overall_readings

This aggregated table (group_and_aggregate) is the source for the overall rainfall bar chart, the summary table, the persisted data file, and the rainfall map layer. The monthly chart, by contrast, works from the row-level (per-event) data — see §4.2.

4. Chart Outputs

4.1 Overall Rainfall Bar Chart

draw_bar_chart (grouped mode) renders total rainfall (mm) per station from the station-aggregated table.

Property	Value
X axis	raingauge_station — one bar per station
Y axis	raingauge_readings (sum) — total mm
Bar colour	#6495ed (cornflower blue)
Plot background	#f5f5f5 (light grey)
Legend	Hidden (showlegend: false)
Bar / group gap	0.05 / 0.05

4.2 Monthly Stacked Bar Chart

draw_bar_chart (stacked mode) renders month-by-month rainfall contributions at each station, using the row-level events data with date components decomposed via decompose_datetime (year, month, month_name).

Calendar ordering

ecoscope_workflows_ext_big_life.tasks.localization.month_names (abbreviated: true) returns the canonical Jan–Dec month-name ordering, passed to stack_order so stacks are sequenced by calendar month regardless of which months have data.

Month colour assignment

apply_color_map maps each time_month_name value to a shade from the **Blues_r** sequential colormap — earlier months are darker blue, later months lighter — giving an intuitive temporal gradient within each stacked bar.

Property	Value
X axis	raingauge_station — one stacked bar per station
Y axis	overall_readings (sum, per month per station)
Stack column	time_month_name
Stack order	Jan → Dec (from month_names task)
Colour column	colors (Blues_r colormap)
Plot background	#f5f5f5
Legend	Visible

4.3 Rainfall-by-Station Summary Table

draw_table renders the station-aggregated table (raingauge_station, raingauge_readings) with sorting and filtering enabled and downloads disabled (enable_download: false). It is persisted as HTML and wired into both the dashboard and the Word report is not applicable here — this widget appears on the dashboard only.

5. Rainfall Map

An interactive DeckGL map combining three static context layers with a clustered layer of rainfall readings, framed automatically to the data extent.

5.1 Land-use Fill Layer

apply_color_map assigns a fill colour to each polygon by its land_use value; create_geojson_layer then renders the filled/stroked polygons at 75% opacity:

Land-use Category	Fill / Line Colour	Opacity
Rangeland	#a39c91	75 %
Rangeland and settlement	#ffa07a	75 %
Settlement and agriculture	#2f4f4f	75 %
Conservancies	#8fbc8b	75 %
National parks and reserves	#4d6600	75 %

5.2 Ranch Boundaries and Electric Fence

The boundaries + fence GeoPackage is filtered on land_use into two separate line layers (no fill):

Feature	Colour	Width / Opacity	Notes
Ranch boundaries	#000000 (black)	0.95 / 55 %	Thin outlines delineating each group ranch
Electric Fence	#2d62b1	5.5 px / 100 %	Bold line marking the electric-fence alignment

5.3 Clustered Rainfall Layer

create_clustered_labeled_scatterplot_layer draws the station readings as clustered circles, sized and labelled by raingauge_readings:

Property	Value	Effect
get_fill_color / get_line_color	[100, 149, 237] #6495ed	Cornflower blue — matches the bar chart colour
get_radius	raingauge_readings	Circle radius scales with the station's total rainfall
radius_units / scale	pixels / 1	Screen-space radius, zoom-independent
cluster_padding	1.0	Spacing applied between clustered points
cluster_max_radius	40 px	Maximum pixel radius used to group nearby stations into a cluster
Text label	raingauge_readings value, monospace, size 15, scaled to the circle radius	Reading is shown directly on each circle
stroked / filled	true / true	Outline and fill both drawn

5.4 View Framing & Layer Composition

envelope_gdf (expansion factor 1.05) computes a bounding envelope around the aggregated station points, and compute_view_state_from_gdf derives an initial camera view (pitch 0, bearing 0, max zoom 15) from that envelope — framing the map to the stations that actually reported data. combine_deckgl_map_layers then composes the land-use, boundary, and fence layers as static base layers with the clustered rainfall layer as a grouped layer, before draw_map renders the final map (max zoom 10, legend placed bottom-right).

6. NDVI Analysis

Vegetation condition is assessed per ranch using the Normalised Difference Vegetation Index (NDVI) derived from MODIS satellite imagery. The analysis produces one historic time-series chart per conservancy, showing how current-period NDVI compares against the multi-year historic range.

6.1 ROI Split by Ranch

split_groups partitions the Amboseli conservancy boundaries GeoDataFrame by the R_NAME column, producing one GeoDataFrame per ranch. Each is used as the roi (region of interest) for a separate calculate_ndvi_range call via mapvalues.

column_first_unique_value extracts the ranch name from each split GeoDataFrame; groupbykey then zips the ranch names with the corresponding NDVI chart HTML so each persisted file is suffixed with the correct ranch name.

6.2 NDVI Method

The user selects one of two MODIS-based NDVI methods at run time via the ndvi_method string variable:

Method	Source	Resolution	Temporal resolution	Best for
MODIS MYD13A1 16-Day Composite	Pre-calculated NDVI	500 m	16-day composites	Phenology studies; quality-filtered best-pixel values; ~0.025 accuracy; may saturate in dense canopies
MODIS MCD43A4 Daily NBAR	NIR/Red band ratio	500 m	Daily (view-angle corrected)	Higher temporal resolution; consistent measurements across acquisitions; more susceptible to cloud gaps

Default is **MODIS MYD13A1 16-Day Composite**. baseline_time_range is currently fixed to null and image_size to a very large fixed value — both marked in spec.yaml as pending support for a user-configurable baseline time range in the web form.

> **Note:** This version of the workflow does not persist a per-ranch NDVI data file (CSV or otherwise) — only the rendered chart (HTML/PNG) is written per ranch.

6.3 Historic Time-series Chart

draw_historic_timeseries renders the per-ranch NDVI chart with three visual layers:

Visual element	Column	Style
Current NDVI line	NDVI	Default plotly line style; labelled 'NDVI'
Historic min-max band	min / max	Filled area between min and max; fill: rgba(144, 238, 144, 0.3) — light green; line: rgba(144, 238, 144, 0.8); labelled 'Historic Min-Max'
Historic mean line	mean	Default plotly line style; labelled 'Historic Mean'

The time axis uses the img_date column. The min-max band is drawn as a filled area between two mode: lines traces, giving a visual envelope of multi-year NDVI variability. When the current NDVI line falls outside this envelope it indicates an anomalously wet or dry condition relative to the historical baseline.

7. Word Report

The final deliverable is a **Word document** (overall_report.docx) assembled from a Dropbox-hosted template populated with every generated chart, the rainfall map, and every per-ranch NDVI chart.

7.1 Template Download

The template file (big_life_weather_report_template.docx) is downloaded automatically from Dropbox at run time (fetch_and_persist_file, up to 3 retries). overwrite_existing: false skips the download on subsequent runs if the file is already cached, saving bandwidth.

7.2 Report Generation

generate_weather_report populates the template context and inserts all map and chart images into the document:

Parameter	Value	Description
template_path	downloaded .docx	Big Life weather report template
output_dir	ECOSCOPE_WORKFLOWS_RESULTS	Output directory
generated_by	"Ecoscope" (fixed string)	Author field in the report — not resolved from the EarthRanger user
box_height / box_width	3.12 in / 5.34 in	Image placeholder dimensions
time_period	analysis time_range	Report period displayed in the header
filename	overall_report.docx	Output filename
validate_images	true	Checks that all expected image files exist before populating placeholders

Images are sourced from the PNG screenshots produced by html_to_png tasks at 2× device scale factor — see §9.3 for the per-task wait-time configuration.

8. Output Files

All outputs are written to the directory defined by ECOSCOPE_WORKFLOWS_RESULTS.

File	Format	Content
overall_rainfall_station_readings	GeoParquet	Total rainfall per station, with point geometry
rainfall_readings_over_time.html / .png	HTML / PNG	Overall rainfall bar chart by station
monthly_rainfall_readings.html / .png	HTML / PNG	Monthly stacked rainfall bar chart, calendar-ordered
rainfall_by_station_summary_table.html	HTML	Sortable, filterable rainfall-by-station table
rainfall_map.html / .png	HTML / PNG	Interactive map: land use, boundaries, fence, rainfall readings
<ranch>.html / .png (one per ranch)	HTML / PNG	Historic NDVI time-series chart per ranch
overall_report.docx	Word	Final populated Big Life weather report

PNGs are rendered at 2× device scale factor for all charts and the map.

9. Workflow Execution Logic

9.1 Skip Conditions

Every task carries two default skip conditions via task-instance-defaults:

- **any_is_empty_df** — if any input DataFrame is empty, the task and its dependants are skipped rather than raising an error.
- **any_dependency_skipped** — if an upstream task was skipped, all downstream tasks that depend on it are also skipped automatically.

The rainfall-event fetch (get_events_data) sets raise_on_empty: true, so EarthRanger returning zero rainfall_rep events for the period produces an explicit error rather than a silent, empty-output run — distinguishing a workflow skip from a genuine data-availability problem.

9.2 mapvalues Fan-out (NDVI per Ranch)

The NDVI pipeline fans out per ranch via mapvalues. split_roi_groups (split_groups) partitions the conservancy GeoDataFrame by R_NAME. Each downstream task in the chain receives the resulting list as its argvalues, processing one ranch per invocation:

Task id	Task	Maps over
calculate_ndvi	calculate_ndvi_range	roi ← split_roi_groups
draw_ndvi	draw_historic_timeseries	dataframe ← calculate_ndvi
get_ranch_name	column_first_unique_value	df ← split_roi_groups
zip_ranch_ndvi	groupbykey	zips draw_ndvi + get_ranch_name
persist_ndvi	persist_text	[text, filename_suffix] ← zip_ranch_ndvi
convrt_ranch_png	html_to_png	html_path ← persist_ndvi

9.3 Screenshot Configuration

Task	wait_for_timeout	max_concurrent_pages	Reason
convrt_ranch_png	10 ms	3	NDVI charts are self-contained HTML; fast render, parallelised
convrt_br_png	10 ms	1	Overall bar chart is self-contained HTML
convrt_monthly_png	10 ms	1	Monthly stacked bar chart is self-contained HTML
convert_map_png	40 000 ms	1	Rainfall map loads external tile layers — long wait ensures tiles render before screenshot

9.4 Dashboard Assembly

gather_dashboard builds the final dashboard (bl_events_dashboard) from the workflow details, time range, and groupers, and exactly four widgets:

- Rainfall Map — the interactive map (§5)
- Raingauge Measurements — the overall rainfall bar chart (§4.1)
- Monthly Rainfall Readings — the monthly stacked bar chart (§4.2)
- Rainfall by Station Summary — the summary table (§4.3)

Per-ranch NDVI charts are *not* added to the dashboard — they are only written as standalone output files and embedded in the Word report.

10. Software Versions

Package	Version	Channel
ecoscope-platform	2.18.0	repo.prefix.dev/ecoscope-workflows
ecoscope-workflows-ext-custom	0.1.0rc14.*	repo.prefix.dev/ecoscope-workflows-custom
ecoscope-workflows-ext-ste	0.0.0rc1.*	repo.prefix.dev/ecoscope-workflows-custom
ecoscope-workflows-ext-mnc	1.0.0.*	repo.prefix.dev/ecoscope-workflows-custom
ecoscope-workflows-ext-wwf-virunga	0.0.0rc9.*	repo.prefix.dev/ecoscope-workflows-custom
ecoscope-workflows-ext-big-life	1.0.0.*	repo.prefix.dev/ecoscope-workflows-custom
pydeck	0.9.2	conda-forge
opentelemetry-sdk	>=1.20.0,<2.0.0	conda-forge

The Big Life-specific tasks (event-detail title mapping, month-name localisation, clustered map layer, weather-report generation) live in ecoscope-workflows-ext-big-life; the bar and historic time-series charts come from ecoscope-workflows-ext-wwf-virunga; and the Dropbox fetch, reprojection, envelope/view-state, and layer combination tasks come from ecoscope-workflows-ext-ste. The runtime environment is managed by **pixi**.